PLASMA RETARDATION IN ALFVÉN'S CRITICAL VELOCITY PHENOMENON

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In an experiment on Alfvén's critical velocity phenomenon, it is observed that the plasma stream retards far more rapidly than can be expected from the theories which relate the retardation mechanism to the processes of ionization of the gas cloud.

Ever since Alfvén proposed the hypothesis of critical velocity for an interaction between a plasma and a neutral gas in relative motion [1], several laboratory experiments with different sets of magnetohydrodynamic parameters have been performed to test it. In the rotating plasma devices, the observed voltage limitation effect is likened to this phenomenon [2–5]. A more direct demonstration and test can be found in the direct plasma–gas collision experiments of Danielsson [6] and Mattoo and Venkataramani [8], where a high velocity plasma stream retards to Alfvén's critical velocity after it passes through a stationary neutral gas cloud in a magnetic field. Most of the experiments have been summarized and discussed in review papers of Danielsson [10] and Sherman [11].

There are many theoretical works based on well-known classical phenomena which can explain the results of rotating devices [12–15]. However, none of these theories can account for the rapid ionization observed in the plasma–gas collision experiments [6–9]. It then becomes necessary to assume some kind of collective interaction for the rapid heating of electrons so that the ionization of the neutral gas can occur rapidly. Also, only theories which are based on collective interaction seem to be directly applicable to Alfvén's original hypothesis for cosmical plasmas [11]. The theories that have been proposed invoke either the formation of space charge [16,17] or two-stream instability [18,19] as mechanisms of collective interaction. Much effort is put into explaining the sharp onset of ionization interaction at the critical velocity and the mechanism of electron heating. On the other hand, these theories give only a broad description of the plasma retardation, and the mechanism of transfer of momentum from the plasma stream to the ionized component of the gas is not clearly enunciated. According to the simple calculation of Danielsson and Brenning [7], the plasma retardation is determined by the processes of ionization. They invoke collective interaction to heat the electrons so that the ionization time, τ, becomes comparable to the transit time of the plasma through the gas. The plasma retardation scale length, L, is related to τ by

$$L = V\tau = \frac{V}{n_g \chi(E)} (M_i/M_n) , \tag{1}$$

where V is the initial plasma velocity, n_g is the gas density, $\chi(E)$ is the ionization function dependent upon the electron energy E, and (M_i/M_n) is the mass ratio of the plasma ion to the neutral gas molecule. Varma [17] has considered the contribution of charge exchange collisions towards plasma deceleration. The purpose of the present experiment is to investigate whether or not ionization processes alone can account for the retardation of the plasma stream motion to the critical velocity.

The experimental parameters are the same as presented earlier [8,9]. The experimental arrangement is shown in fig. 1. The whole system is under a high vacuum of 10^{-6} Torr. A transverse magnetic field of

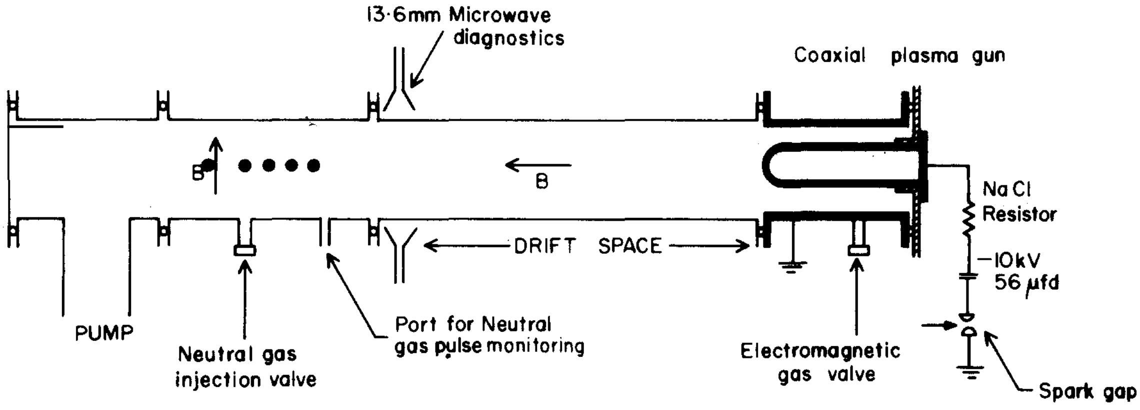

Fig. 1. Schematic of the experiment. The black circles indicate the locations of $V \times B$ probes for measuring plasma stream velocity at different points in the gas cloud.

1000 G is provided in the interaction region where a gas cloud is injected by an electromagnetic value. The dynamics of the cloud is monitored by reflex discharge probes [20]. The plasma stream is generated by a coaxial plasma gun and accelerated into a drift tube along a magnetic field. The plasma density as measured by 22 GHz interferometer is $\sim 5 \times 10^{18}$ m^{-3}: For the electron temperature, the Langmuir probe indicates that it is ~5 eV. The plasma velocity can be varied between 2 and 100 km s^{-1} and the corresponding maximum kinetic energy of the plasma ions varies from 50 eV for hydrogen to 2 keV for argon. The opening of the gas valve and the electric discharge in the gun are synchronized so that the gas cloud in the interaction region has the desired axial spread and average density. The average gas density can be varied between 10^{17} and 10^{21} m^{-3}. The density scale length (e-folding length) of the gas cloud varies from 1 cm at the average gas density of 10^{21} m^{-3} to 50 cm at 2×10^{17} m^{-3}. In the measurements reported here, the scale length of the gas cloud is ensured to be greater than the observed velocity retardation scale length. Since diffusion time of the neutral gas cloud is ~100 μs and transit time of the plasma stream through neutral gas cloud is only a few μs, the gas cloud appears stationary to the penetrating stream The shot-to-shot variation in the density of the gas cloud is less than 10%. The plasma retardation is observed by measuring plasma velocity with $V \times B$ probes placed at different points in the gas cloud; the distance between the probes varies from minimum of 0.8 cm to maximum of 8 cm. As reported

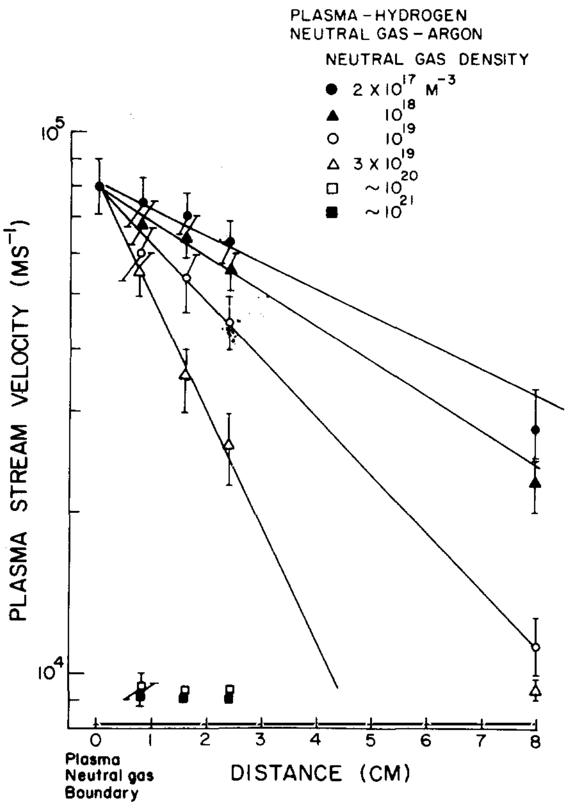

Fig. 2. Hydrogen plasma velocity at different points in the argon gas cloud of indicated gas density. The plasma–neutral gas boundary is the position in the experimental region upto which the plasma moves unretarded. Alfvén's critical velocity of argon is 8.7 km s^{-1}.

probes are in good agreement with those obtaines by measuring time-of-flight between two spaced Langmuir probes.

Fig. 2 shows the variation of the velocity of hydrogen plasma as it streams through the argon neutral gas cloud at different gas densities. The velocity decreases, rapidly at high gas densities, as the plasma penetrates into the neutral gas. At the last $V \times B$ probe, the velocity is lower by an order of magnitude in the dense clouds compared with the same in the absence of neutral gas. Defining velocity retardation scale length as the e-folding distance, we have tabulated the observed retardation lengths at different gas densities in table 1. This table also shows the observed retardation length in the cases where oxygen, nitrogen, neon and argon plasma interact with the argon gas cloud. The numbers in brackets in this table are the computed values of the retardation length using eq. (1). These calculations assume the plasma electrons are heated to a temperature of about 100 eV during the onset of the plasma–neutral gas interaction. This assumption is fairly valid in view of the measurement of electron temperature in the experiment of Danielsson [6], and Danielsson and Brenning [7]. These measurements indicate that the heating of electrons takes place at the penetration of the plasma into the neutral gas cloud and that the electron temperature may be as high as 100 eV. It must be remarked here that, in our experiment, the assumed electron temperature may be too high in case of hydrogen plasma interacting with any neutral gas cloud. This is because the maximum initial ion energy is ~50 eV which is less than the assumed electron temperature. It earlier [8,9], the velocity measurements by these is for this reason that we have not given the computed values of L in tables 1 and 3 for the interactions where plasma stream is of hydrogen.

The results summarized in table 1 clearly indicate that the retardation scale length does not scale as (M_i/M_n). At any average neutral gas density, we do not observe the expected increase in scale length by a factor of 40 as the mass of plasma ion is varied from 1 hydrogen to 40 for argon. The maximum increase is by a factor of 3.5 at 2×10^{17} m^{-3} gas density. This is verified further by studies in which mass ratios are inverted by interchanging the gases supplied to the plasma gun and the gas cloud region. These results are displayed in table 2. A comparison between tables 1 and 2 shows that althought the observed plasma retardation is always slower when $M_i < M_n$ than when $M_i > M_n$, the scale length for direct and inverted mass ratio are not related to each other by $(M_i/M_n)^2$. As a matter of fact, the ratio of scale length for $M_i > M_n$ to the same for the inverted mass ratio ranges between 3 and 5 seems to be independent of mass ratio. Further, even after normalizing the measured scale lengths for the ionization function $\chi(E)$, they do not show the expected scaling.

The variation of scale length with the gas cloud density is such that it increases as the gas density decreases. This variation is more pronounced for interactions which have $M_i \gtrsim M_n$. The observed scale length, however, does not increase by an order of magnitude if the

Table 1
Retardation scale lengths in cm, of indicated plasma going through argon gas cloud $M_i \lesssim M_n$.

Plasma	n (m^{-3})			
	10^{20}	10^{19}	10^{18}	2×10^{17}
Hydrogen	<0.5	4.5	7.0	10.0
Nitrogen	<0.5	2.0	4.0	7.0
	(0.14)	(1.4)	(14)	(70)
Oxygen	<0.5	2.5	3.0	10.0
	(0.16)	(1.6)	(16)	(80)
Neon	<0.5	2.0	10.0	30.0
	(0.2)	(2)	(20)	(100)
Argon	<0.5	4.0	7.0	35.0
	(0.4)	(4)	(40)	(200)

Table 2
Retardation scale lengths in cm, for argon plasma going through indicated neutral gases $M_i \gtrsim M_n$.

Gas	n (m^{-3})			
	10^{20}	10^{19}	10^{18}	2×10^{17}
Hydrogen	4.0	19.0	24.0	36.0
	(80)	(800)	(8000)	(40000)
Nitrogen	1.5	6.0	12.0	20.0
	(1.4)	(14)	(140)	(700)
Oxygen	2.0	10.0	15.0	40.0
	(22.5)	(225)	(2250)	(11250)
Neon	1.0	5.0	13.0	30.0
	(3.6)	(36)	(360)	(800)
Argon	<0.5	4.0	7.0	35.0
	(0.4)	(4)	(40)	(200)

gas density is reduced by the same magnitude. An empirical fit shows that $L \propto n_g^{-\alpha}$, where α ranges from 0.15 for non-symmetrical ($M_i \neq M_n$) to 0.5 for symmetrical ($M_i = M_n$) plasma–neutral gas interactions.

We notice from table 1 that there is a reasonably good agreement between the experimentally observed and the theoretically expected values of the plasma retardation length at $n_g = 10^{19}\ m^{-3}$. But this agreement is not seen at lower gas densities, $n_g < 10^{19}\ m^{-3}$, and even at $n_g = 10^{19}\ m^{-3}$ for those interactions which have $M_i > M_n$ (see table 2). In fact, the experimentally observed value of the retardation length is an order of magnitude less than the expected value for some of the interactions listed in tables 1 and 2. In order to make sure that this result does not suffer from any ambiguity due to the dependence of the retardation length on M_i/M_n, we have studied interactions in which mass ratio = 1. In these interactions the same gas was used for the formation of the plasma stream and the gas cloud. The results are displayed in table 3. It again shows that the computed scale length is an order of magnitude greater than the observed value and confirms that the plasma retards faster than what is expected from eq. (1).

The discrepancy between the observed and the expected retardation scale lengths can be reduced, at the most, by a factor of two if the value of V in eq. (1) corresponds to the average velocity of the plasma during the interaction time rather than the initial velocity. The agreement can be made excellent if only the the value of V were just about the critical velocity of the neutral gas. But, at this velocity, the rapid ionization

Table 3
Retardation scale lengths in cm, for symmetric plasma–neutral gas interactions $M_i = M_n$.

Plasma/Gas	n (m^{-3})			
	10^{20}	10^{19}	10^{18}	2×10^{17}
Hydrogen	1.6	4.0	9.0	20.0
Nitrogen	1.2	3.0	7.0	25.0
	(0.5)	(5)	(50)	(250)
Oxygen	2.0	5.0	15.0	40.0
	(9)	(90)	(900)	(4500)
Neon	1.0	4.5	12.0	26.0
	(1.8)	(18)	(180)	(900)
Argon	<0.5	4.0	7.0	35.0
	(0.4)	(4)	(40)	(200)

interaction, associated with Alfvén's critical velocity phenomenon, cannot occur and the ionization function $\chi(E)$ has a considerably small value. Thus, it cannot be conjectured that the presumption of equaling V to the initial plasma velocity is the cause of the discrepancy reported here. Further, it may be argued that the assumed electron temperature for the calculation of $\chi(E)$ is either too high or too low. This can be ruled out for two reasons. Firstly, the experimental observations of Danielsson and Brenning [7] have established that electrons can get heated to 100 eV. Secondly, the ionization function, $\chi(E)$, has a broad maximum in the range of 70–100 eV for all the gases used in the formation of the gas cloud in our experiment. Even if the electrons were heated to more than the presumed value, it would not have changed the value of $\chi(E)$ very much. Also, assuming electron temperature less than 100 eV would only accentuate the discrepancy further as the value of $\chi(E)$ is lower than that at 100 eV. Hence, the discrepancy cannot either be explained on the basis of the ionization function.

A comparison between the measured and the calculated interaction times also leads to the same conclusions as derived from the measurements of the retardation scale length. We find that the observed deceleration time is less than the calculated ionization time for 100 eV electrons et all the gas densities considered here; the difference increases as n_g decreases.

In the calculation of loss of momentum suffered by the plasma during the interaction time through binary encounters between the plasma ions and gas particles, we have considered ion–neutral (elastic, resonant and non-resonant charge transfer, and ionization) and ion–ion Coulomb collisions. Of all the types of collisions, resonant charge transfer collisions are the most important at high gas densities.

To estimate the momentum loss due to asymmetric charge exchange collisions we have taken into account a functional dependence of the charge exchange cross section on the velocity of the plasma stream because of deceleration of the stream during its penetration in the gas cloud. Using published data of cross section [22–28], the estimated loss of momentum is not more than 15% of initial momentum of the plasma stream at $n_g = 10^{19}\ m^{-3}$. For a calculation of the momentum loss due to resonant charge exchange collisions, we have assumed that all the newly formed ions are accelerated to Alfvén's critical velocity in a single step before they undergo charge exchange collisions. At $n_g = 10^{19}$ m^{-3}, additional momentum loss due to resonant charge exchange collisions is about 25% of the initial momentum in the plasma stream. Since the resonant charge exchange cross section is quite high, it may be argued that if newly formed ions are decelerated by resonant charge exchange with the neutral gas they must be re-accelerated by the plasma in the same way as newly formed ions. Such a process would add to further braking of the plasma motion. It should be borne in mind, however, that the plasma deceleration length and correspondingly the acceleration length for the newly formed ions is much smaller than the resonant charge exchange mean free path. This will allow only a small fraction (~10%) of the newly formed ions to undergo the processes of reacceleration. Hence the net effect of reacceleration is not so much to decelerate the plasma but to give a spread in the velocity component of the newly formed ions along the direction of the stream. At $n_g = 10^{19}$ m^{-3} additional momentum losses due to all the collision processes other than due to charge transfer collisions amount to about 10%. Thus, it seems that binary collisions can partially account for the observed plasma retardation at high gas densities and therefore may explain the difference between the observed and the expected retardation lengths in some of the interactions studied in our experiment. However, the difficulty of explaining the observed fast retardation at low gas densities ($n_g = 10^{17} - 10^{18}$ m^{-3}) cannot be resolved even by invoking these collisions. This is because the role of these collisions is considerably reduced at these gas densities. Our calculations indicate that the plasma may not lose more than 1% of its initial momentum in the interaction time of 10μs through resonant charge transfer collisions at $n_g = 2 \times 10^{17}$ m^{-3}. The contribution from all other collisions totals to about 4% at this gas density.

We have also considered the momentum losses to the walls. Our estimate indicates that these losses are not significant enough to explain the difference between the observed and computed plasma retardation lengths. Further, our experimental results present evidences for viewing that the momentum losses to the walls are negligibly small to merit any serious considerations. Firstly, the plasma retardation in the interactions where the plasma ion gyroradius, r_i, is greater than the diameter of the device D, is not observed to be faster than those for which $r_i < D$. Secondly, α seems to have the same value (~0.5) for all symmetric interactions such as A–A, N–N, O–O and H–H. If the momentum losses to the walls were of any importance, then for the A–A interaction they should have been very much different from those for H–H.

The conclusion of our studies is that a plasma stream can partially lose its momentum/energy through binary collisions only when it is interacting with the dense gas cloud ($n_g > 10^{19}$ m^{-3}). In order to explain the observed fast plasma retardation in the diffused gas cloud ($n_g < 10^{19}$ m^{-3}) there is a need for invoking a far more efficient momentum loss mechanism than binary collisions. On the other hand, it is not clear either from theories or from the observations of plasma gas collision experiments whether a part of the momentum of the plasma is lost in accelerating the ionized component of the gas to the Alfvén's critical velocity or in heating of the plasma and newly created ions. In this context, we wish to emphasize that Alfvén's orig-

Table 4
Reaction rates for charge exchange processes.

(a) resonant

Plasma–neutral gas	$\langle \sigma V \rangle$ a) (10^{-14} m^3 s^{-1})	$\langle \sigma V \rangle$ b) (10^{-14} m^3 s^{-1})
H–H	3.1	2.45
N–N	0.82	0.57
O–O	4.0	1.02
Ne–Ne	0.9	0.43
Ar–Ar	0.75	0.22

(b) asymmetric

Plasma–neutral gas	H–Ar	N–Ar	O–Ar	Ne–Ar	Ar–H	Ar–N	Ar–O	Ar–Ne
$\langle \sigma V \rangle$ a) (10^{-15} m^3 s^{-1})	0.2	0.28	0.45	1.0	0.74	0.7	0.95	0.82

a) Averaged over the velocity range of initial plasma velocity to Alfvén's critical velocity.
b) V corresponds to Alfvén's critical velocity of the neutral gas.

inal hypothesis does not explicitly imply that the ionized component of the gas has to be accelerated to the critical velocity simultaneously with the deceleration of the plasma to the critical velocity. In fact, if one considers the effect due to reaccelerations of stationary ions resulting from successive resonant charge-exchange collisions between the accelerated newly formed ions and the neutral gas atoms, it is very obvious that the ionized component of the gas will have a dispersion in its velocity component along the direction of the plasma stream. Thus, the ionized component may appear to have been anisotropically heated. This apparent ion heating has to be differentiated from the one that would result due to the modified two-stream instability [19,21]. This has to be verified from experimental observations.

As mentioned earlier, the theories based on collective interaction either do not adequately explain the mechanism of plasma retardation and the acceleration of the ionized gas [16,18,19] or predict a plasma retardation according to eq. (1) which, as shown by our experimental observations, is a far slower process. There is a clear need of improving the theoretical models which will give a better description of the plasma–neutral gas interaction experiments. The exchange of momentum between different regions of plasma and therefore transport of particles, momentum and heat content of electrons may be some of the effects which present theories have not included.

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